

PLCA vs CSMA/CD on one 10BASE-T1S segment, repeated

Generated by `make_access_report.py` from `access_20261006_114158.json`.

Bench. ESP-A: ESP32-S3 + LAN8651 MAC-PHY, PLCA coordinator (id 0 of 2), SPI 20.00 MHz. Converter: LAN8670 + LAN9355, dial id 1. ESP-B: ESP32-S3 + W5500 on the converter's 100BASE-TX port, in the PC's place. 1000 B UDP. The same five phases as the 2026-10-01 PC comparison; every number below is per round, mean $\pm$ 95 % CI (Student t) over rounds.

CSMA/CD rounds pending. The converter's PLCA is a DIP switch. With it on and only the node switched to `csma`, the converter does not fall back to CSMA/CD: it stops transmitting (every probe and every flood datagram from ESP-B lost, `access_20261006_113637`). The CSMA/CD rounds need DIP 1 off by hand; this report fills its second column when they exist.

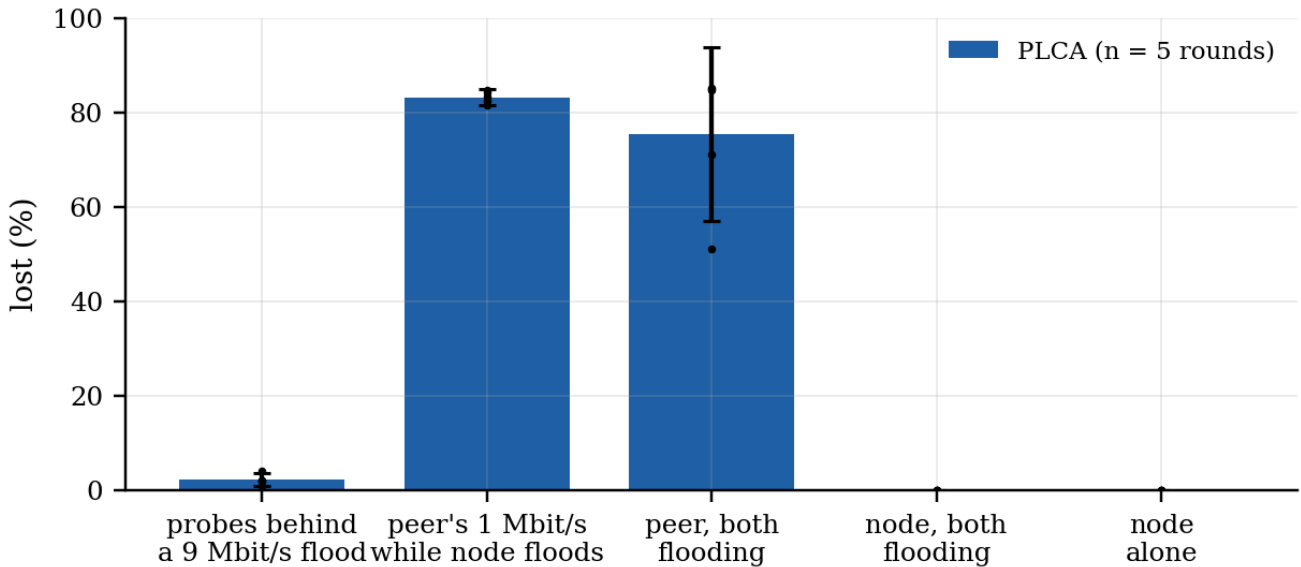


Fig. A1 — Loss in each phase, per method; bars are means, whiskers 95 % CI, dots single rounds.

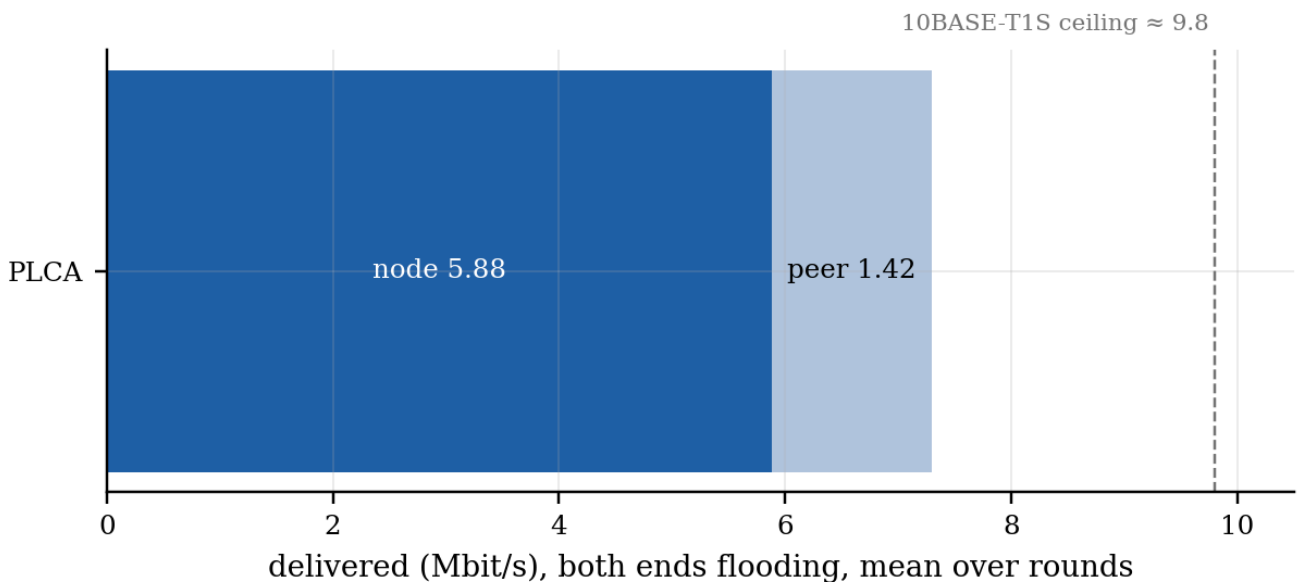


Fig. A2 — How the bus is shared when both ends flood: the node's delivered rate (solid) and the peer's (light).

phase	metric	PLCA (n = 5)
idle	64 B round trip, median (from ESP-B)	2.59 ± 0.01 ms (2.58–2.60)
idle	probes lost	0.0 ± 0.0 % (0.0–0.0)
peer floods 9 Mbit/s	node's probes lost	2.2 ± 1.4 % (1.0–4.0)
peer floods 9 Mbit/s	node's probe round trip, median	3.04 ± 0.19 ms (2.86–3.23)
peer floods 9 Mbit/s	node's probe round trip, p99	1730.18 ± 1041.21 ms (821.54–2639.82)
peer floods 9 Mbit/s	flood delivered to the node	8.52 ± 0.02 Mbit/s (8.51–8.54)
peer floods 9 Mbit/s	flood lost	5.3 ± 0.2 % (5.1–5.5)
node floods, peer 1 Mbit/s	node -> peer delivered	7.05 ± 0.01 Mbit/s (7.04–7.06)
node floods, peer 1 Mbit/s	node -> peer lost	0.00 ± 0.00 % (0.00–0.00)
node floods, peer 1 Mbit/s	peer -> node delivered (of 1.0)	0.17 ± 0.02 Mbit/s (0.15–0.18)
node floods, peer 1 Mbit/s	peer -> node lost	83.2 ± 1.7 % (81.4–84.6)
both flood	node -> peer delivered	5.88 ± 1.10 Mbit/s (4.49–6.48)
both flood	node -> peer lost	0.00 ± 0.00 % (0.00–0.00)
both flood	node's longest gap	527 ± 977 ms (7–1782)
both flood	peer -> node delivered	1.42 ± 1.30 Mbit/s (0.73–3.13)
both flood	peer -> node lost	75.4 ± 18.4 % (51.0–85.1)
node alone	node -> peer delivered	7.22 ± 0.01 Mbit/s (7.22–7.23)
node alone	node -> peer lost	0.00 ± 0.00 % (0.00–0.00)
node alone	arrival gap p99	2.50 ± 0.00 ms (2.50–2.50)

Reading it (PLCA). The node never lost a datagram it sent: 0.00 % while it flooded alone, 0.00 % with the peer sending, 0.00 % with both flooding. Its small probes got onto a bus the peer was flooding at 9 Mbit/s with 2.2 % loss (median 3.04 ms), but their tail is in seconds (p99 1.73 s; the converter carried 8.52 of the 9 offered). It is not a queue of excess offered load: offering 7.5–9.0 Mbit/s (`tail_vs_offered_poll.json`, 2026-10-07) leaves tails of 0.4–4.4 s at every rate, worst below the converter's drain rate, so the replies are held up inside the flooding board itself (its echo and its paced blast share one W5500 and one CPU) – the median hides it. What PLCA does not protect is the converter's own stream: while the node floods, 83 % of the peer's 1 Mbit/s is lost, and with both flooding 75 % — the converter's two-way limitation the vendor lists (issue 4.3), seen in every round. When both flood, the node keeps 5.88 Mbit/s but stalls for up to 527 ms (mean of per-round maxima): it waits for transmit opportunities rather than losing frames.

Limits. Two nodes on the bus, one of them a converter whose transmit path is the known weak point; rounds of one method run back to back (the DIP is set by hand), so slow drift between the two blocks is not randomised out; one bus length (~1 m).